

## Module 1 | Class 1 Assignment: ML Type Classification Exercise

### Task 1: Classify 17 Real-World Scenarios

| No | Scenario                                                                                                                     | Learning Type              | Justification                                                               | Input                                                                  | Output                                                |
|----|------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------|
| 1  | Uztelecom predicts which customers will cancel their contract next month.                                                    | Supervised (labelled).     | There is the label that is customers.                                       | Customer data: salary, cash flow, history, payment behavior            | Prediction: will cancel/ will not cancel              |
| 2  | A self driving car learns to navigate Tashkent traffic by repeatedly trying routes and receiving penalties for unsafe moves. | Reinforcement.             | Checking for penalties through traffic patterns.                            | Tashkent traffic data, routes, actions                                 | Predicting unsafe moves                               |
| 3  | Uzum marketplace recommends products to users based on their browsing and purchase history.                                  | Supervised                 | The label is products.                                                      | Products data, purchase history, customers' browsing                   | Predicting future products for customers' interesting |
| 4  | A system groups Tashkent neighborhoods by internet usage patterns without any predefined categories.                         | Unsupervised (Clustering). | No labels, there is not clear label that the system what groups to find it. | Internet usage data about citizens' online hours, speed, consumed data | Groups with similar citizens according input          |
| 5  | An email provider classifies incoming messages as spam or not spam.                                                          | Supervised.                | It learns with labelled data as spam or not spam                            | Incoming messages data                                                 | Classifying spam or not spam                          |

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|----|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------|
| 6  | A robot arm in a Navoi mining facility learns to sort ore by trial and error, getting a reward signal for correct picks.                         | Reinforcement. | The robot arm learns through trial or error to get a reward signal for correct picks. | Sensor data, arm movements                                 | Correct ore sorting actions               |
| 7  | Dynamic pricing for Uztelecom mobile data plans — adjusting prices in real time based on demand signals and user response.                       | Reinforcement. | Adjusts prices in real-time based on user response (reward = revenue/profit)          | Demand signals and user response data.                     | Optimal price at each moment              |
| 8  | A hospital in Samarkand uses patient records to predict diabetes risk.                                                                           | Supervised.    | Has labeled data (has diabetes / no diabetes)                                         | Patients' records as data                                  | Predicting diabetes risks                 |
| 9  | An algorithm segments Uztelecom's 10 million subscribers into marketing groups based on usage behavior — no one told it how many groups to make. | Unsupervised   | No predefined labels or number of groups (Clustering)                                 | A dataset about the subscribers' usage behavior            | Segmenting groups based on usage behavior |
| 10 | A chess engine improves its play by competing against itself over millions of games.                                                             | Reinforcement  | Learns by playing against itself and getting rewards for winning                      | Board state, moves                                         | Better chess moves                        |
| 11 | Detecting SIM card fraud at Uztelecom by flagging transactions that look different from a user's normal pattern.                                 | Unsupervised   | Machine should detect SIM card fraud. There is no clearly SIM card fraud              | A dataset about flagging transactions                      | Flagged suspicious transactions           |
| 12 | A news aggregator clusters articles into topics automatically.                                                                                   | Unsupervised   | No labels, automatically finds topics (Clustering)                                    | Articles                                                   | clusters articles into topics             |
| 13 | Predicting house prices in Tashkent from square footage, district, and floor number.                                                             | Supervised     | House prices is as a labelled data                                                    | A dataset about square footage, district, and floor number | Predicting house prices                   |

|    |                                                                                                                       |               |                                                                                                 |                                                   |                                                 |
|----|-----------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------|
| 14 | A warehouse robot learns the fastest route between shelves by exploring and receiving time-based penalties.           | Reinforcement | Machine need learn the fastest route according to exploring and receiving time-based penalties. | A dataset about the fastest route between shelves | Optimal route                                   |
| 15 | A model translates Uzbek text to English using a dataset of sentence pairs.                                           | Supervised    | The label is Uzbek text, it is clear for machine                                                | A dataset of sentence pairs                       | Predicting English text according to Uzbek text |
| 16 | Spotify creates a “Discover Weekly” playlist by finding users with similar listening habits and cross-recommending.   | Unsupervised  | Uses collaborative filtering to find patterns (no direct labels)                                | Listening history                                 | Predicting a “Discover Weekly” playlist         |
| 17 | An algorithm detects unusual spikes in Uztelecom’s network traffic without labeled examples of “normal” vs “anomaly.” | Unsupervised  | There is no clear label as examples of “normal” vs “anomaly”                                    | Uztelecom’s network traffic data                  | Predicting unusual spikes                       |

## Task 2: Reflection Questions

### Questions:

1. Which scenarios were hardest to classify? What made them tricky?
2. Can a single business problem use more than one type of ML at different stages? Give an example.
3. If you had to pick one ML type that is most useful for Uzbekistan's telecom industry right now, which would it be and why?

### Answers:

1. The hardest scenarios for me to classify were #7 (Dynamic Pricing), #9 (Customer Segmentation), #10 (Chess Engine), and #16 (Spotify Discover Weekly). These were tricky because they sit in a gray area between different types of machine learning. For example, Dynamic Pricing can look like Supervised Learning because it uses historical data, but it also learns in real-time from user reactions, which is closer to Reinforcement Learning. Similarly, Spotify's recommendation system and customer segmentation don't have clear "correct answer" labels, making it difficult to decide between Supervised and Unsupervised approaches. These ambiguous cases taught me that real-world problems are often not purely one type of ML.
2. Yes, a single business problem can definitely use more than one type of machine learning at different stages of the solution. A good example is customer churn prediction for Uztelecom. First, we can use Unsupervised Learning (clustering) to segment customers into different groups based on their usage behavior. Then, for each segment, we can apply Supervised Learning to predict which customers are likely to leave next month. This multi-stage approach usually gives better and more accurate results than using only one type of ML. In real projects, companies often combine Unsupervised, Supervised, and even Reinforcement Learning to solve complex problems.
3. If I had to pick only one, I would choose Supervised Learning. The telecom industry in Uzbekistan (especially companies like Uztelecom) has many important problems that can be solved effectively with labeled data, such as predicting customer churn, detecting fraud, classifying spam calls/messages, and predicting network failures. Supervised Learning is

the most practical and immediately useful because there is already a lot of historical labeled data available from billing systems and customer records. While Unsupervised and Reinforcement Learning are also valuable, Supervised models can deliver faster business impact and are easier to measure (accuracy, precision, recall), which is very important for companies in our market right now.